INCH-POUND

MIL-D-70631A (AR)
AMENDMENT 2
24 March 1992
SUPERSEDING
AMENDMENT 1
10 April 1989

MILITARY SPECIFICATION

Dispenser, Mine: M138 (FLIPPER)

This amendment forms a part of Military Specification MIL-D-70631A (AR), dated 15 May 1987, and is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1.1 Specifications and Standards, under "SPECIFICATIONS", delete "FEDFERAL" and substitute "FEDERAL".

PAGE 2

* 2.1.1 Specifications and Standards, under "STANDARDS", add:

"MIL-STD-171, Finishing of Metal and Wood Surfaces"

* 2.1.2 Other Government documents, drawings and publications; delete drawing "9399999, Chemical Agent Resistant Coatings (CARC)".

PAGE 4

* Add the following paragraph:

"3.3.7 Durability Test. The dispenser shall launch a minimum of 3960 M79 Practice Mines (11 missions) without equipment failure when operated as specified in 4.5.1.7."

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PAGE 5

3.6.1 Welding: Change second sentence to read: "All welds shall be classified Class A, unless otherwise specified".

* 3.7 Protective finish; delete "Dwg. 9399999" and replace with "MIL-STD-171."

PAGE 6

4.3.1 Submission, Change drawing number from "P/N 9399924-1" to "P/N 9359924-1".

PAGE 10

* 4.4.2.1 Clamp, Machined, Category 206, delete in its entirety and substitute:

"206, Groove to hole designation incorrect (2 places), 0.65%, 3.2, Visual".

PAGE 15

4.4.2.6 Hub, Change inspection method for Major 105 from "Visual" to "Gage".

PAGE 16

4.4.2.7 Tire Assembly, Add Major 105:

"105, Tire inflation incorrect, 0.40% 3.2, Gage".

PAGE 18

* 4.4.2.8 Axle, Major 121, delete footnote indicator "1/" and paragraph reference "4.5.4". Also delete footnote "1/" in its entirety from Notes block.

PAGE 19

4.4.2.9 Flywheel, change word "concentricity" to "True Position" in the following categories (4 places): Majors 102, 104, 106 and 109.

PAGE 21

* 4.4.2.10 Wheel Half, delete Major 111 in its entirety and substitute:

"111, Thickness from inner top rim to bottom surface incorrect, 100%, 3.2, Gage".

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* 4.4.2.14 Spacer, Pintle, Major 104, delete footnote indicator "1/" and paragraph reference "4.5.4". Also delete footnote "1/" in its entirety from Notes block.

PAGE 28

* 4.4.2.15 Pintle, Major 116, delete footnote indicator "1/" and paragraph reference "4.5.4". Also delete footnote "1/" in its entirety from Notes block.

PAGE 33

* 4.4.2.20 Mounting Plate, Switch, Proximity, Major 110, delete "Gage" and substitute "Visual".

PAGE 34

4.4.2.21 Raceway, Change category from "Minor" to "Major".

* 4.4.2.21, Raceway, delete Major 103 in its entirety.

Major 106, delete in its entirety and substitute:

"106, Raceway top mounting surface not parallel to Datum C, 0.40%, 3.2, Gage".

PAGE 35

* 4.4.2.21 Raceway, Major 115, delete "(8 places)" and substitute "(7 places)".

PAGE 36

* 4.4.2.21, Raceway, Major 127, delete in its entirety and substitute:

"127, Distance to short and long support mounting holes incorrect (2 places), 0.40%, 3.2, Gage".

* 4.4.2.21 Raceway, Add the following Major inspections:

"134, Distance to bottom edge of transmitter cutout incorrect, 0.40%, 3.2, Gage".

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"135, Distance to top edge of transmitter cutout incorrect, 0.40%, 3.2, Gage".

"136, Distance to inner edge of transmitter cutout incorrect, 0.40%, 3.2, Gage".

"137, Distance to outer edge of transmitter cutout incorrect, 0.40%, 3.2, Gage".

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* 4.4.2.22 Top, Raceway, Major 101, delete "(8 places)" and substitute "(7 places)".

Major 102, delete paragraph reference "4.5.4" in its entirety.

Major 106, delete in its entirety.

Major 107, Add "Angular" before the word "Location".

PAGE 38

* 4.4.2.22 Top Raceway, Major 115, delete "(8 places)" and substitute "(7 places)".

Add the following Major inspections:

"117, Radius of front bearing assembly mount cutout incorrect, 0.40%, 3.2, Gage".

"118, Distance to outer edge of transmitter cavity incorrect, 0.40%, 3.2, Gage".

"119, Distance from centerline to bottom of transmittor cavity incorrect, 0.40%, 3.2, Gage".

PAGE 40

* 4.4.2.24 Bushing, Sleeve, Major 108, delete footnote indicator "1/" and delete paragraph reference "4.5.4". Also delete Note "1/" in Notes block in its entirety.

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PAGE 41

4.4.2.25 Plate, Retaining, Bearing, delete Major 109 in its entirety and substitute:

"109, Perpendicularity of bearing hole incorrect, 0.40%, 3.2, Gage".

PAGE 42

* 4.4.2.25 Plate, Retaining, Bearing, Minor 205, delete in its entirety and substitute:

"205, Location of support(s) incorrect (2 places), 0.65%, 3.2, Gage".

PAGE 43

* 4.4.2.25 Plate, Retaining, Bearing, delete Minor 206 in its entirety.

PAGE 47

4.4.2.28 Base Assembly, delete Major 115 in its entirety and substitute:

"115, Perpendicularity of bearing hole incorrect, 0.40%, 3.2, Gage".

PAGE 50

* 4.4.2.30 Shelf Assembly, Major 106, delete footnote indicator "1/" and Paragraph Reference "4.5.4".

PAGE 51

4.4.2.30 Shelf Assembly, delete Minor 205 in its entirety.

PAGE 55

* 4.4.2.33 Launcher Assembly, Title Block, delete "Sheet 2 of 3" and substitute "Sheet 2 of 2".

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PAGE 57

* 4.4.2.34 Dispenser, Mine, Functional Testing, renumber Majors "110 thru 118" to "114 thru 122".

PAGE 58

* 4.4.2.35 Lock, Adapter, Major 106, delete footnote indicator "1/" and delete paragraph reference "4.5.4". Also delete Note "1/" in Notes block in its entirety.

PAGE 64

* 4.5.1 Arming and dispensing. Add the following NOTE after paragraph:

"NOTE: If, during testing, the dispenser is restrained or restricted from normal movement by mechanical means, the restraining devices(s) shall simulate the shock absorber action of a human operator."

4.5.1.3 Cold Temperature, line 2, delete "Method 502.1" and substitute "Method 502.2".

PAGE 65

* 4.5.1.5 Transportation vibration, delete, paragraph in its entirety and substitute:

"4.5.1.5 Transportation vibration. The dispenser shall be conditioned in accordance with MIL-STD-810, Method 514.3, category 8, test procedure I, figure 514.3-7 thru 514.3-9 (500 miles) and figure 514.3-10 thru 514.3-22 (48 miles), except for the following changes:

FIGURE	FREQUENCY	PSD VALUE
514.3-7	5	.032
514.3-7	6	.005
514.3-7	7	.088
514.3-7	8	.135
514.3-8	5	.032
514.3-9	5	.032
514.3-9	6	.055
514.3-9	7	.088
514.3-9	8	.135

The dispenser shall be tested after conditioning as specified in 4.5.1.".

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* 4.5.1.7 Add the following paragraph:

"Durability Test. This test to be performed at ambient temperature. The Clamp Assembly (Dwg. 9359850) shall be securely mounted to a test stand or vehicle. The M138 Dispenser shall be assembled to the Clamp Assembly and secured with the pin and chain assembly. A 26 + 1 VDC power source shall be connected to the dispenser using the Cable Assembly (Dwg. 9360017).

Prior to initial launching of M79 practice mines, dispenser shall launch one(1) Test Mine (P/N 9359924-1) at normal SD setting and one(1) Test Mine (P/N 9359924-2) at extended SD setting. The test mines shall be launched into a drop cloth, erected for soft catch, and tested for proper voltage outputs as specified in 4.5.2.

After launching and testing the two test mines, the dispenser shall launch 720 M79 practice mines (2 missions). (The dispenser may be stopped between missions to allow the motor to cool.) Subsequently, the two test mines shall again be launched and tested.

The test mines shall be dispensed and tested after every two launching missions of 720 M79 practice mines until all 3960 M79 mines are launched, after which the two test mines shall again be launched and tested.

Upon completion of testing, the dispenser shall be examined for excessive wear including condition of the Friction Lining (P/N 9359807), the underside of the Top, Raceway (P/N 9359906) and the Tire (P/N 9359817)".

* 4.5.2.1 Extended setting, static test, Subparagraph "a", add "one(1) sine wave generator", after "required:". Subparagraph "d", delete in its entirety and substitute:

"d. Set the sine wave generator to 10 volts peak-to-peak at 1200 Hz. Connect the generator to TB3-11(+) and TB3-1(-). Turn on power to the electronics box and set the mode switch to "extended". Slowly increase the generator frequency until the ready light comes on."

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* 4.5.2.1 Extended setting, static test, Subparagraph "h", first sentence, delete "35 volts ± 3" and substitute "31 volts ± 3V".

PAGE 67

* 4.5.5 Humidity indicator test, delete last sentence in its entirety and substitute:

"The indicator card shall indicate a maximum relative humidity of thirty(30) percent for the shipment to be allowed to proceed".

PAGE 68

* 6.3 Submission of inspection equipment designs for approval; delete "ATTN: AMSMC-QAT-I (D)" and replace with "ATTN: SMCAR-QAT-I".

* 6.4 Distribution of ammunition data cards and test data; delete "ATTN: AMSMC-QAT-M (D)" and replace with "ATTN: SMCAR-QAT-M".

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